KPMG

GLOBAL FRAUD DOCUMENT DETECTION (GFDD) STANDARD OF PROCEDURE

Version 1.4

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1. Introduction

1.1 Purpose

The Standard Operating Procedure (SOP) provides comprehensive guidance on the operation and utilization of the Global Fraud Document Detection (GFDD) tools. Designed for users tasked with verifying document authenticity and detecting fraudulent activities, this SOP serves as a critical resource for understanding the functionalities and capabilities of the GFDD system.

It outlines detailed procedures for conducting analyses on a variety of document types, including leveraging metadata, pixel statistics, principal components, and AI modification checks. By defining the operational framework, this SOP ensures users can effectively harness GFDD tools to enhance document integrity and reliability.

1.2 Scope

The GFDD system encompasses an integrated suite of tools and methodologies that collectively address the challenges of document fraud detection. The key components include:

1. **AI Integrity Check:** This component underscores the importance of detecting AI-generated modifications within visual content, evaluating the likelihood of AI-driven alterations or creations that threaten document authenticity. By analyzing intricate patterns and inconsistencies, this tool safeguards document integrity.
2. **Pixel & AI Analysis:** Through pixel-level examination and AI interaction insights, this module provides a thorough analysis of documents to identify anomalies. These detected anomalies serve as vital indicators of potential tampering, helping users mitigate risks associated with fraudulent activities.
3. **Fraud Visual Tools:** Employing advanced visual investigative techniques, this component facilitates the detection of document manipulations through methodologies such as edge detection, error level analysis, and copy-move forgery. These tools vividly highlight suspect areas within documents, allowing users to conduct detailed assessments.
4. **Text Consistency Checker:** Designed to verify the integrity and coherence of textual data, this tool evaluates documents for consistency and accuracy. By identifying discrepancies in text content, it provides crucial insights into potential fraud, supporting efforts to ensure document precision and truthfulness.

This structured approach enables users to apply comprehensive analytical processes across various document types—ensuring effective detection and mitigation of fraudulent activities, thus reinforcing document authenticity.

1.3 Input Requirements

PARAMETERS	DETAILS
INPUT TYPES	Bank Statements, ID cards, invoices, receipts, claims, etc.
SUPPORTED FORMATS	Images and Documents (See table below)
PROCESSING MODE	Single image or batch mode (20 files max per batch, 10MB per file)

Supporting Image Format	Document Format
1. Windows bitmaps - *.bmp, *.dib 2. GIF files - *.gif 3. JPEG files - *.jpeg, *.jpg, *.jpe 4. JPEG 2000 files - *.jp2 5. Portable Network Graphics - *.png 6. WebP - *.webp 7. AVIF - *.avif 8. Portable image format - *.pbm, *.pgm, *.ppm, *.pxm, *.pnm 9. PFM files - *.pfm 10. Sun rasters - *.sr, *.ras 11. TIFF files - *.tiff, *.tif 12. OpenEXR Image files - *.exr 13. Radiance HDR - *.hdr, *.pic	1. Portable Document Format - *.pdf

The Global Fraud Document Detection (GFDD) tools represent a pivotal innovation in fraud detection, providing robust methodologies and meticulous procedures for analyzing document authenticity. This SOP elucidates the processes and configurations needed to maximize the potential of these tools, empowering users to make informed decisions regarding document integrity.

2. AI Integrity Check

2.1 Objective

The AI Integrity Check is designed to evaluate the authenticity of visual content, such as images, collections of images, or videos, by determining whether they have been generated or modified using artificial intelligence (AI). This sophisticated tool utilizes advanced machine learning models to scrutinize inherent characteristics and unique patterns within the visuals, enabling precise classification and providing vital insights into the originality of the data.

2.2 Purpose

The primary purpose of the AI Integrity Check is to detect and identify components within documents—such as photos, logos, and text—that have been altered using AI technology. This functionality ensures the authenticity of the content by safeguarding against AI-driven manipulations.

2.3 Key Features

- **Advanced Algorithm Utilization:** The AI Integrity Check employs sophisticated AI algorithms to meticulously assess and validate document authenticity.
- **Probability-Based Alerts:** It generates alerts based on the likelihood of alterations, with a threshold set to trigger when the probability of real content is below 75%.

2.4 Outputs

The tool delivers comprehensive evidence and probability scores, detailing the extent and likelihood of AI modifications. This output equips users with crucial data for further analysis and decision-making regarding document integrity.

2.5 Configuration Procedures

1. Image Selection
    - Upload Image: Users can upload either a single image/PDF or multiple files for analysis to verify potential AI-generated elements.

2. **Run the Analysis**
   - Process Command: Initiate the analysis using the 'Process Image' button, employing the Pixel and AI Analytical Suite.
3. **Review Results**
   - Upon completion, results are displayed on-screen, including a summary indicating whether the document is considered to have fraudulence and the reasons for such classification.
   - Results can be exported in CSV format for further examination and record-keeping.

2.6 Detection Results

Results are presented in a structured format to support user understanding and interpretation of findings. Example results include:

Sample Analysis Results:

Filename	Human Probability	AI Probability	Alert	Alert Message
HKID.JPG	87.14%	12.86%	0	None
Indo ID.jpg	2.91%	97.09%	1	Indo ID.jpg has 97.09% probability AI-generated.
US Driving Licence.JPG	90.96%	9.04%	0	None
HKID_2.JPG	80.19%	19.81%	0	None

2.7 Interpretation of Results

1. **Human Probability**
   - Indicates the likelihood that the image or document was generated by a human, offering insight into its authenticity.
2. **AI Probability**
   - Displays the probability that the image or document was generated or altered by AI, flagging potential modifications.
3. Alert
   - Alerts are triggered when Human Probability falls below 75%, classifying the document as likely AI-generated and suggesting a higher chance of manipulation.

By adhering to these analysis settings, users can effectively leverage the AI Integrity Check to evaluate visual data for AI modifications, ensuring comprehensive evaluation and robust fraud detection within the GFDD system.

2.8 Use Case Examples

Pending snapshots

3. Pixel & AI Analysis

3.1 Objective

The Pixel & AI Analysis suite provides a powerful, integrated approach for document examination, employing multiple analytical methods to uncover document authenticity with precision and confidence. This comprehensive toolset applies methodologies such as metadata analysis, pixel evaluation, principal component analysis (PCA), and AI detection to ensure thorough fraud detection and document validation.

3.2 Purpose

The suite integrates these diverse methodologies to perform a multidimensional examination of documents, focusing on pixel analysis, PCA, and AI detection, thus providing a thorough and reliable assessment of document integrity.

3.3 Key Features

- **Comprehensive Analysis:** Combines statistical techniques, including metadata analysis, pixel intensity distribution, and principal components, to detect variances and identify anomalies.
- **Proactive Detection:** Generates alerts and detailed summaries on fraud probability and patterns, enabling users to take proactive measures against possible fraudulent activities.

3.4 Output

Produces detailed data files with alert conditions and comprehensive metrics, offering insights into potentially fraudulent activities. This allows for meticulous evaluation of document authenticity and integrity.

3.5 Configuration Procedures

1. **Image Selection**
   - Users can upload either a single image/PDF or a batch of up to 20 files for analysis. The tool processes all uploaded items, generating detailed results.
2. **Enable Inclusive Mode for Pixel Statistics Analysis**
   - **Inclusive Mode Checkbox:** Check this option to include additional statistical measures such as standard deviation and distribution in the analysis.
3. **Enable Equalize for Principal Component Analysis**
   - **Equalize Functionality:** Equalizes the histogram of the projected values, aiding in normalizing distribution and highlighting anomalies.
4. **Run the Analysis**
   - **Process Command:** Initiate the analysis using the 'Process Image' button.
5. **Review Results**
   - Assess on-screen results after analysis, including the number of alerted documents and specific alert messages for each algorithm.
   - A summary indicates whether a document is considered fraudulent, supplemented by reasons for potential fraud identification.
   - Results can be exported in a CSV format for thorough review.

3.6 Detection Logic

An image is classified as Potential Fraud based on the following logic:

Metadata Analysis

- **Examine:** Creation/modification dates and editing tool metadata.
- **Alert Conditions:** Triggered if modification date is later than the creation date or if editing tools are detected.
- **Fraud Classification:** An alert from Metadata Analysis signifies potential fraud.

Pixel Statistics

- **Analyze:** Pixel intensity distributions for anomalies (e.g., color dominance, sharpness issues).
- **Alert Conditions:** Triggered if three or more specified anomalies are detected.
- **Fraud Classification:** Co-occurrence with PCA alerts indicates potential fraud.

Principal Component Analysis (PCA)

- **Evaluate:** Number of channels, directions, and patterns for abnormalities.
- **Alert Conditions:** Triggered if two or more abnormal geometries or projections are detected.
- **Fraud Classification:** Alerts support fraud suspicion.

AI Detection

- **Classify:** Using machine learning to assess authenticity.
- **Alert Conditions:** Triggered if AI-generated probability exceeds 25%.
- **Fraud Classification:** Alerts indicate potential fraud.

Algorithm	What to Check	Possible alert messages	Conditions to trigger alert	Conditions to classify as Potential Fraud
Metadata Analysis	Examine creation/modification dates and editing tool metadata	**1.** The last edit date is after when it was created. **2.** Editing software was detected.	If either alert is triggered, the image fails Metadata Analysis.	Alert from Metadata analysis classifies as Potential Fraud
Pixel Statistics	Analyze pixel intensity distributions	Multiple pixel anomalies (e.g., color dominance, sharpness issue) **1.** Parts of the image are too dark or missing colors. **2.** Colors differ more than expected. **3.** One color dominates the image. **4.** The image appears flat without depth. **5.** The image seems unusually noisy. **6.** The image might be sharper than normal. **7.** The image is excessively sharp. **8.** Multiple patterns or changes are detected.	If 3 or more alerts messages have been triggered, the image fails Pixel Statistics.	Triggering Pixel Statistics and PCA alerts together indicates Potential Fraud
PCA	Evaluate principal components and detect abnormal variance or vector directions	**1.** The color balance seems off, check for edits **2.** One color channel is overemphasized. **3.** There is a high contrast difference. **4.** Unusual data pattern detected, possible tampering **5.** Possible signs that content is fake or altered. **6.** Some parts may be copied and pasted or artificially made.	If 2 or alerts messages have been triggered, the image fails PCA.	
AI Detection	Classify image authenticity using machine learning	More than 25% likelihood that the image was created by AI.	Alert triggered when AI-generated probability exceeds threshold	Alert triggered by AI detection will instant Potential Fraud

3.7 Overview of Algorithms

3.7.1 Metadata Analysis

Objective

Metadata Analysis is devised to scrutinize file properties such as creation and modification dates, editing tools usage, and camera specifications to provide insights into the image's history and integrity. Through metadata evaluation, potential abnormalities that indicate tampering or manipulation can be identified.

Methodology

- **Property Examination**
  - Extraction and analysis of metadata fields to uncover critical information about an image's lifecycle.

Details

- **Creation Date and Modification Date**
  - Evaluates whether modification dates later than creation imply post-creation edits.
- **Editing Tools**
  - Detection of editing software signatures signaling probable alterations.

Alert Triggering Conditions

Alerts occur when modification dates surpass creation dates or editing tools are detected.

- **The last edit date is after when it was created**: Indicates that the image has been edited after initial creation, especially concerning raw files.
- **Editing software was detected**: Presence of metadata showing usage of editing software identifies potential alterations.

Output

Provides reports on metadata anomalies offering evidence of potential tampering.

3.7.2 Pixel Statistics Analysis

Objective

Pixel Statistics Analysis evaluates pixel intensity distributions within images, employing statistical measures (e.g., minimum, average, maximum) to uncover anomalies or inconsistencies that may suggest manipulation.

Methodology

- **Mode Selection and Statistical Evaluation:**
  - Analyzes pixel values using selected modes with options for additional metrics like standard deviation.

Details

- **Pixel Intensity Metrics**
  - Assesses pixel values across different modes to detect sudden spikes or bimodal distribution patterns signaling manipulation.

Alert Triggering Conditions

Alerts arise due to deviations in standard deviation, abnormal intensity, or outliers, indicating possible tampering.

- Sudden Spike in Standard Deviation: Indicates unusual variations in pixel intensities suggesting tampering.
- Abnormal Average Pixel Intensity: Significant deviation from expected values suggests alterations in image parts.
- Bimodal Distribution: If Indicates manipulation of different regions within the image.
- Outliers in Minimum or Maximum Values: Suggests tampering by extreme value deviations.

Output

Generates detailed summaries of pixel anomalies contributing to document validation.

3.7.3 PCA Analysis

Objective

PCA projects pixel values onto principal components to identify patterns and anomalies, enabling the detection of unusual variations in pixel values deviating from expected norms.

Methodology

- **Component Projection and Variation Detection**
  - Analyzes pixel projections to spot unexpected patterns or color shifts indicating tampering.

Details

- **Components and Anomalies**
  - Evaluates eigenvalues and eigenvectors to detect unusual variance or directionality.

Alert Triggering Conditions

Alerts are generated by atypical eigenvalues, unexpected eigenvector directions, and mean vector anomalies.

- **High Variance in Principal Components:** Unusual eigenvalues indicate manipulation or altered regions.
- **Unusual Eigenvector Directions:** Unexpected directions suggest image manipulation or tampering.
- **Mean Vector Anomalies:** Abnormal mean values indicate potential alterations suggesting manipulation.

Output

Provides variance and vector direction maps highlighting PCA-derived anomalies.

3.7.4 AI Detection

Objective

AI Detection employs machine learning models to classify images based on inherent characteristics, determining whether visual content is generated or altered by artificial intelligence.

Methodology

- **AI-driven Classification**
  - Assesses image authenticity using models trained on features indicative of AI influence.

Details

- **Model-Based Prediction**
  - Utilizes probability assessments to determine the presence of AI-generated elements.

Alert Triggering Conditions

- Alerts trigger when human probability is below 75%, suggesting AI generation.

Output

Reports are generated with probability scores and identifications of AI influence, enhancing fraud detection.

3.8 Detection Results

Presented in a structured format to aid user comprehension. Sample results include:

Bank_Statement_F.jpg:

Test	Pass / Fail	Alert Messages	Potential Fraud
Metadata Test	Pass	N/A	No
Pixel Test	Pass	N/A	
PCA Test	Pass	N/A	
AI Test	Pass	N/A	

AI and real.pdf:

Test	Pass / Fail	Alert Messages	Potential Fraud
Metadata Test	Pass	N/A	Yes
Pixel Test	Fail	• Parts of the image are too dark or missing colors. • One color dominates the image. • The image appears flat without depth.	
PCA Test	Pass	N/A	
AI Test	Fail	• 50.62% generated by AI	

AI generated Image.jpg:

Test	Pass / Fail	Alert Message	Potential Fraud
Metadata Test	Pass	N/A	Yes
Pixel Test	Fail	• Parts of the image are too dark or missing colors. • Multiple patterns or changes are detected. • Colors differ more than expected. • One color dominates the image.	
PCA Test	Fail	• There is a high contrast difference.	

		• The color balance seems off, check for edit	
AI Test	Fail	• 97.09% generated by AI	

3.9 Interpretation of Results

In the Pixel & AI Analysis suite, alerts are triggered based on specific criteria during the document analysis. While users do not see the detailed checks themselves, they are presented with summarized and classified alerts to interpret the document's status effectively.

1. **Summary of Alerts**
   - Users receive a summary indicating the number of documents flagged for alerts based on internal analyses. This summary includes a decision on whether the document is classified as fraudulent, along with the reasons for potential fraud identification.
2. **Condition for Alert Triggering and Potential Fraud Classification**
   - **Metadata Analysis:** An alert is triggered if anomalies in creation or modification dates are detected, or if there is evidence of editing tool usage, leading to a classification as potential fraud.
   - **Pixel Statistics:** Alerts are activated if multiple pixel anomalies are detected, such as abnormal intensity or distribution variations. When combined with PCA alerts, these may classify the document as potential fraud.
   - **Principal Component Analysis (PCA):** Anomalies such as signs of manipulation, high variance in principal components, or unexpected eigenvector directions prompt additional alerting. Combined with other alerts, this may indicate potential fraud.
   - **AI Detection:** If AI-generated content detection exceeds set thresholds (more than 25% likelihood), an immediate alert is triggered, categorizing the document as potential fraud.

This combined interpretation empowers users to understand and act upon the analysis results effectively, ensuring comprehensive fraud detection.

3.10 Use Case Examples

Pending Latest snapshots

This detailed description ensures an understanding of the Pixel & AI Analysis processes, ensuring meticulous examination and detection of potential document manipulations within the GFDD system.

4. Fraud Visual Tools

4.1 Objective

The Fraud Visual Tools suite is designed to provide a comprehensive visual assessment of documents, enabling the detection of potential tampering and fraudulent modifications. These tools employ advanced visual and analytical techniques to facilitate thorough analysis and informed fraud detection.

4.2 Methodology

Technique Utilization

- Utilizes methodologies such as edge detection, error level analysis (ELA), and copy-move forgery detection to provide clear visual representations of suspected alterations. This visual approach ensures that users can effectively identify, pinpoint, and analyze areas of concern within documents.

4.3 Key Features

The Fraud Visual Tools provide a comprehensive set of functionalities for visually evaluating documents and identifying potential signs of tampering or manipulation. These tools harness advanced image analysis techniques to highlight alterations through detailed visual representations, supporting effective fraud detection strategies.

Edge Detection

- **Purpose:** Captures abrupt changes in luminance or image intensity, revealing possible signs of tampering.
- **Feature Details:** Utilizes optimized algorithms to enhance edge clarity, providing clear visual cues for boundary assessment within images.

Error Level Analysis (ELA)

- **Purpose:** Highlights discrepancies in image quality following compression, aiding in the detection of alterations.
- **Feature Details:** Employs comparison of original and recompressed images to expose inconsistencies, with adjustable parameters for sensitivity calibration.

Copy-Move Forgery Analysis

- Purpose: Examines documents for duplicated areas that could indicate manipulation or fraudulent content.
- **Feature Details:** Uses key point detection and clustering algorithms to find and visualize suspected forgery clusters, enabling in-depth analysis of duplications.

User-Friendly Interface

- The tools are equipped with intuitive interfaces allowing users to adjust analysis settings and view results efficiently.
- Export functionality is available, enabling users to download visual reports for archival or further examination.

4.4 Algorithm Logic

1. **Visual Representation:**
    - Processes images to create visual output that highlights potential tampered regions through techniques like edge mapping, error level visualization, and detection of copy-move operations.
2. **Comparative Assessment:**
    - Evaluates visual evidence against expected norms and historical baselines to identify deviations suggestive of manipulation.

4.5 Output

The Fraud Visual Tools generate detailed visual representations of regions suspected of alteration, providing:

- **Visual Evidence:** Offers graphical data aiding in the further analysis and validation of document integrity.
- **Insights and Indicators:** Supplies actionable insights into potential tampering, enabling users to conduct a meticulous examination and achieve precise fraud identification.

4.6 Performance and Optimization

Resource Utilization

- The website handles user interactions and data processing efficiency without specific hardware requirements, leveraging cloud-based services and APIs where applicable.

These technical requirements establish the foundational software and operational components necessary for maintaining and running the GFDD system effectively, enabling continuous fraud detection services to users via its website interface.

4.7 Edge Detection

4.7.1 Summary

Edge Detection is a critical functionality designed to identify points or pixels in an image where there are significant changes in luminance (intensity). These points form line segments known as edges, which are essential for detecting boundaries and contours within an image. This process aids in identifying potential areas of tampering or manipulation by highlighting abrupt changes.

4.7.2 Configuration Options and Procedures

1. **Image Selection**
   - **Upload Image:** Allows users to upload single or multiple images or PDFs for analysis.
2. **Radius (Detect Sensitivity)**
   - **Values:** 1 – 15 pixels
   - **Description:** Controls sensitivity; smaller radii detect finer details; larger radii detect broader edges.
   - **Configuration:** Set the radius based on desired sensitivity; default of 3 pixels is common. Use 1-4 pixels for fine details and 10-15 for broader edges.
3. **Contrast (Edge Visibility Enhancement)**
   - **Values:** 0% to 100%
   - **Description:** Adjusts contrast to enhance edge visibility; higher values make edges more prominent.
   - **Configuration:** Adjust contrast to achieve desired visibility; start with 50% and increase to 70-100% for more prominence. Reduce if edges appear too sharp or noisy.
4. **Grayscale Output Checkbox**
   - **Description:** Converts output to grayscale, simplifying edge visualization and analysis.
   - **Configuration:** Check this box to simplify visualization and focus on edges.
5. **Apply Filter Button**
   - **Description:** Applies the configured edge detection settings to the image.
   - **Configuration:** Click 'Apply Filter' to implement the settings on the image.

4.7.3 Interpretation of Results

1. **Clear Edges**
   - Well-defined edges indicate effective configuration and proper detection.
2. **Noise Reduction**
   - Ensure the image is not overly noisy or cluttered; adjust radius and contrast for clarity.
3. **Visibility**
   - Make sure edges are visible enough with appropriate contrast adjustments.

By applying these analysis settings, users can effectively utilize Edge Detection to reveal important features and identify potential tampering in images. This ensures thorough image analysis and evaluation.

4.8 Error Level Analysis

4.8.1 Summary

A technique used to detect alterations in digital images by analyzing the quality differences at various compression levels. When an image is compressed and then recompressed, the quality of the image can vary across different regions. ELA highlights these variations, which can indicate areas of tampering or manipulation.

4.8.2 Configuration Options and Procedures

1. **Image Selection**
   - Upload Image: Allows users to upload single or multiple images or PDFs for analysis.
2. **Quality (JPEG Compression Quality)**
   - **Values:** 1% to 100%.
   - **Description:** Determines the JPEG compression quality level for image analysis. Lower quality levels, such as 1-20%, increase sensitivity to compression artifacts.
   - **Configuration:** Set the preferred quality level for image analysis at various compression levels; starting with a lower value enhances sensitivity.
3. **Scale (Output Scaling Factor)**
   - **Values:** 1% to 100%.
   - **Description:** Determines the JPEG compression quality level for image analysis. Lower quality levels, such as 1-20%, increase sensitivity to compression artifacts.
   - **Configuration:** Set the preferred quality level for image analysis at various compression levels; starting with a lower value enhances sensitivity
4. **Linear Checkbox**
   - **Description:** Enables linear interpolation during image scaling, improving visual quality.
   - **Configuration:** Enable this checkbox to apply linear interpolation.
5. **Greyscale Checkbox**
   - **Description:** Converts the output to grayscale, simplifying visualization and focusing on intensity differences.
   - **Configuration:** Check this box to convert output to grayscale for easier visualization.
6. **Contrast (Error visibility enhancement)**
   - **Value:** 0% to 100%
   - **Description:** Enhances visibility of compression artifacts by adjusting contrast, with higher values making artifacts more pronounced.
   - **Configuration:** Adjust contrast to accentuate compression artifacts; starting with a moderate value, such as 50%, is advisable.

4.8.3 Interpretation of Results

1. **High Intensity Colors**
   - Areas with high intensity colors (e.g., red) indicate significant compression artifacts, potentially suggesting tampering or manipulation.
2. **Low Intensity Colors**
   - Areas with low intensity colors (e.g., blue) denote minimal artifacts, indicating unaltered or original regions.
3. **Patterns and Anomalies**

- Observe consistent patterns or anomalies in color intensity that may reveal specific areas of tampering.

By following these analysis settings, users can effectively employ Error Level Analysis to detect and evaluate possible manipulations in images, ensuring thorough and reliable forensic analysis.

4.9 Copy Move Forgery

4.9.1 Summary

Copy-Move Forgery Detection is a sophisticated technique designed to identify areas within an image where portions have been copied and pasted, often to conceal or obscure vital features. The tool employs advanced key point detection and matching algorithms to systematically locate duplicated regions, providing comprehensive insights into potential fraudulent alterations.

4.9.2 Configuration Options and Procedures

1. **Image Selection**
   - **Upload Image:** Allows users to upload single or multiple images or PDFs for analysis.
   - **Mask On/Off Toggle:** Option to use a mask image to focus analysis on specific regions. When turned on, users can upload a binary mask image where white pixels indicate regions to be analyzed.
2. **Detection Parameters**
   - **Detector Algorithm:** Selects the key point detection algorithm, with options including:
   - **BRISK (Binary Robust Invariant Scalable Key points):** Efficient binary descriptor for textured images, robust to illumination changes.
   - **ORB (Oriented FAST and Rotated BRIEF):** Fast, general-purpose detector combining FAST key point detection and BRIEF descriptor.
   - **AKAZE (Accelerated-KAZE):** Enhanced algorithm suited for complex textures and varying lighting.
   - **Response Threshold Percentage:** Configurable from 0% to 100% to set the sensitivity of key point detection, with higher values focusing on fewer, robust key points.
   - **Minimum Match Percentage for Forgery Detection:** Sets the percentage of matches required to consider a region as forgery, ensuring high confidence.
3. **Advanced Options**
   - **Distance Threshold for Key Point Comparison:** Adjustable from 1 to 100, setting strictness for matching key points.
   - **Minimum Number of Points Forming a Valid Forgery Cluster:** Configurable between 1 and 20, determining how many key points are needed to form a forgery cluster.
4. **Display Options**
   - **Hide Lines:** Checkbox to toggle visibility of lines connecting matched key points.
   - **Show Key Points:** Checkbox to toggle visibility of detected key points.

4.9.3 Interpretation of Results

The tool presents detection results in a structured format for user clarity:

1. **Key Points**
   - Total detected key points within the image.
2. **Filtered Key Points**
   - Number of key points remaining after applying response thresholds.
3. **Matches**
   - Total key point matches identified between regions.

4. **Clusters**
    - Number of valid forgery clusters based on configured settings.
5. **Regions**
    - Distinct areas suspected of copy-move forgery.

By leveraging these configuration options, detection parameters, and advanced visualization tools, users can effectively identify and interpret areas of potential fraud within documents and images, ensuring detailed and reliable forensic analysis.

4.10 Use Case Examples

Pending Snapshots

This suite delivers powerful capabilities for uncovering signs of tampering, ensuring a meticulous examination of document authenticity through advanced visual techniques.

5. Text Consistency Checker

5.1 Objective

The Text Consistency Checker employs advanced Optical Character Recognition (OCR) technology to meticulously analyze and verify text within documents. By focusing on ensuring consistency and accuracy, this tool is especially efficient for examining financial documents like bank statements, verifying important details such as cash flow entries and transaction dates against known data.

5.2 Methodology

OCR Implementation

- Utilizes Tesseract OCR, an open-source text recognition engine, configured to support multi-language text recognition for comprehensive analysis across diverse documents.

Page Segmentation Modes (PSM)

- PSM 3: Full page segmentation without OSD for standard document layouts.
- PSM 6: Assumes a single, uniform text block, ideal for tables and structured text.
- PSM 11: Optimized for sparse text commonly seen in invoices with scattered fields.

OCR Engine Modes (OEM)

- OEM 0: Traditional Tesseract OCR, favoring processing speed.
- OEM 1: LSTM-based neural network for superior accuracy.
- OEM 2: Combined mode using both traditional and LSTM engines.
- OEM 3: Auto-selects the best engine for optimal results.

Automated Text Extraction:

- Employs image preprocessing, such as CLAHE and resolution optimization, to enhance text recognition accuracy.

5.3 Purpose

Harnesses Optical Character Recognition (OCR) to support text-based verification with a focus on financial documents, ensuring data integrity.

5.4 Key Features

Automated Consistency Verification:

- Extracts and scrutinizes text for consistency across documents, identifying anomalies in contexts such as cash flow entries and transaction dates.

Data Comparison:

- Compares extracted text against verified data, ensuring integrity validation.

5.5 Output

Generates comprehensive reports detailing inconsistencies, providing valuable insights necessary for further analysis and fraud detection.

5.6 Configuration Procedures

1. **Image Selection**
   - Users can upload single documents or batches (up to 20 files) for analysis in formats like PDFs and images.
2. **OCR Settings Configuration**
   - **Page Segmentation Mode (PSM):** Users can select the appropriate PSM, such as:
     - PSM 3: Fully automated page segmentation.
     - PSM 6: Treat as a single text block for structured content.
     - PSM 11: For sparse text with dispersed fields.
   - **OCR Engine Mode (OEM):** Choose the OCR engine mode for optimal performance:
     - OEM 0: Classical engine for speed.
     - OEM 1: LSTM-based neural network for accuracy.
     - OEM 2: Combined mode for balanced performance.
     - OEM 3: Automatically selects the most suitable engine.
3. **Language Selection**
   - Users can configure the OCR for multiple languages, such as English, Chinese (simplified or traditional), or others as needed.
4. **Template Selection for Bank Statements**
   - Select the bank's statement template, providing preset configurations tailored for different financial institutions to enhance accuracy.
5. **Transaction Cashflow Replication**
   - Enable replication checks to confirm consistency in transaction balances or amounts within documents.
6. **Date Alignment Verification**
   - Verify statement periods to ensure all transaction dates fall within predefined ranges.
7. **Description Accuracy Audit (Optional)**
   - Users must upload a mapping list, containing valid account numbers or names, to validate transaction descriptions against provided data.
8. **Run Analysis**
   - Execution Command: Click the 'Process Document' button to begin text consistency checks.
9. **Review Results**
   - Examine on-screen results post-analysis, detailing transaction success rates, alignment, and description accuracy. Export results to CSV for a comprehensive review.

5.7 Detection Logic

Algorithm	What to Check	Possible alert messages	Conditions to trigger alert	Conditions to classify as Potential Fraud
Cashflow Consistency Check	Verify transaction balances and amounts	Success Rate less than 90%	Any success rate below threshold in statements	Alert triggered will instantly classify as Potential Fraud
		Inconsistent amounts in other documents	Any inconsistencies in amounts for other documents	
Date Alignment Verification	Confirm transaction dates	Dates outside statement period for statements	Any inconsistent dates in other documents	Alert triggered will instantly classify as Potential Fraud
	Ensure date consistency in other documents	Inconsistent dates in other documents	Any description mismatches in statements	
Description Accuracy Audit (Optional)	Validate transaction descriptions	Mismatches in descriptions for statements	Any description mismatches in statements	Alert triggered will instantly classify as Potential Fraud
	Verify detail accuracy in other documents	Inconsistent details in other documents	Any inconsistent names/details with client info in other documents	

5.8 Overview of Algorithms

The Text Consistency Checker utilizes a suite of algorithms to ensure the integrity and coherence of textual data within financial and other document types. Each algorithm focuses on identifying specific types of inconsistencies that could indicate potential fraud.

5.8.1 Cashflow Consistency Checker

Objective

The Cashflow Consistency Check algorithm is designed to simulate and verify the alignment of cash flows within bank statements. This ensures that each transaction aligns with starting and ending balances, verifying the consistency and accuracy of financial data.

Methodology

- Adjusts balance calculations for each transaction, assessing any discrepancies that arise.
- Displays the success rate of successful versus failed replications, allowing users to evaluate potential imbalances.

Alert Triggering Conditions

- **Replication Success Rate Below 90%:** Triggers alerts indicating potential fraud due to imbalances in bank statements.
- **Inconsistent Amounts in Other Documents:** Alerts activate when amounts in documents like medical invoices don't align as expected.

5.8.2 Date Alignment Verification

Objective

- The Date Alignment Verification algorithm confirms that all transactions within a bank statement fall within the specified statement period, detecting any date-related discrepancies.

Methodology

- Evaluates transaction dates compared to predefined statement periods; anomalies suggest potentially fraudulent activities.

Alert Triggering Conditions

- **Transaction Dates Outside Statement Period:** Flags transactions falling outside the specified period in bank statements.
- **Inconsistent Dates in Other Documents:** Alerts are generated when dates in documents like medical invoices do not conform to expected timelines.

5.8.3 Description Accuracy Audit (Optional)

Objective

- This optional component evaluates the logical consistency of transaction descriptions in bank statements, emphasizing accurate account details and counterparty names.

Methodology

- Compares transaction descriptions to trusted mappings (e.g., valid account numbers, names) to identify discrepancies.

Functionality Requirements

- User-provided mapping lists are necessary for validation against known criteria.

Alert Triggering Conditions

- **Mismatch in Transaction Description:** Alerts trigger due to disparities in account details or counterparty names in bank statements.
- **Inconsistent Details in Other Documents:** Alerts occur when documents, such as medical invoices, contain unmatched information compared to the client-provided lists.

5.9 Detection Results

Presented in a structured format to aid user comprehension. Sample results include:

Bank Statement 202502.jpg:

Test	Pass / Fail	Alert Messages	Potential Fraud
Cashflow Consistency Checker	Pass	N/A	No
Date Alignment Verification	Pass	N/A	

Bank Statement F.jpg:

Test	Pass / Fail	Alert Messages	Potential Fraud
Cashflow Consistency Checker	Pass	N/A	Yes
Date Alignment Verification	Fail	Transaction Dates Outside Statement Period	

AI generated Image.jpg:

Test	Pass / Fail	Alert Message	Potential Fraud

Cashflow Consistency Checker	Fail	Replication Success Rate Below 90%	Yes
Date Alignment Verification	Pass	N/A	
Description Accuracy Audit	Fail	Mismatch in Transaction Description	

5.10 Interpretation of Results

1. **Transaction Cashflow Replication**
   - **Outcome:** Success rate percentage for cashflow replication; alerts trigger below 90% indicating inconsistencies.
   - **Fraud Classification:** Alerts signify possible fraud due to discrepancies.
2. **Date Alignment Verification**
   - Outcome: Validates dates against specified periods, alerting for anomalies.
   - **Fraud Classification:** Alerts indicate documents as potentially fraudulent.
3. **Description Accuracy Audit (Optional)**
   - **Outcome:** Evaluates logical consistency against uploaded mapping lists, triggering alerts for mismatches.
   - **Fraud Classification:** Discrepancies categorize documents as potential fraud.

These settings and interpretations allow users to methodically verify text consistency, facilitating effective fraud detection and ensuring document integrity.

5.11 Use Case Examples

Pending snapshots

These settings and interpretations allow users to methodically verify text consistency, facilitating effective fraud detection and ensuring document integrity.